Post-training: Instruction Tuning, Alignment, and Test-Time Compute

Why Post-training Matters

- Base LLMs are trained only to predict the next token
- They often ignore instructions, hallucinate, or generate harmful content

Post-training makes models more helpful, safe, and controllable

Two Core Problems of Base Models

- Not sufficiently helpful: fail to follow user instructions
- Not sufficiently safe: may produce harmful, false, or toxic outputs

Post-training aligns model behavior with human goals

Instruction Tuning Compared to Other Finetuning Methods

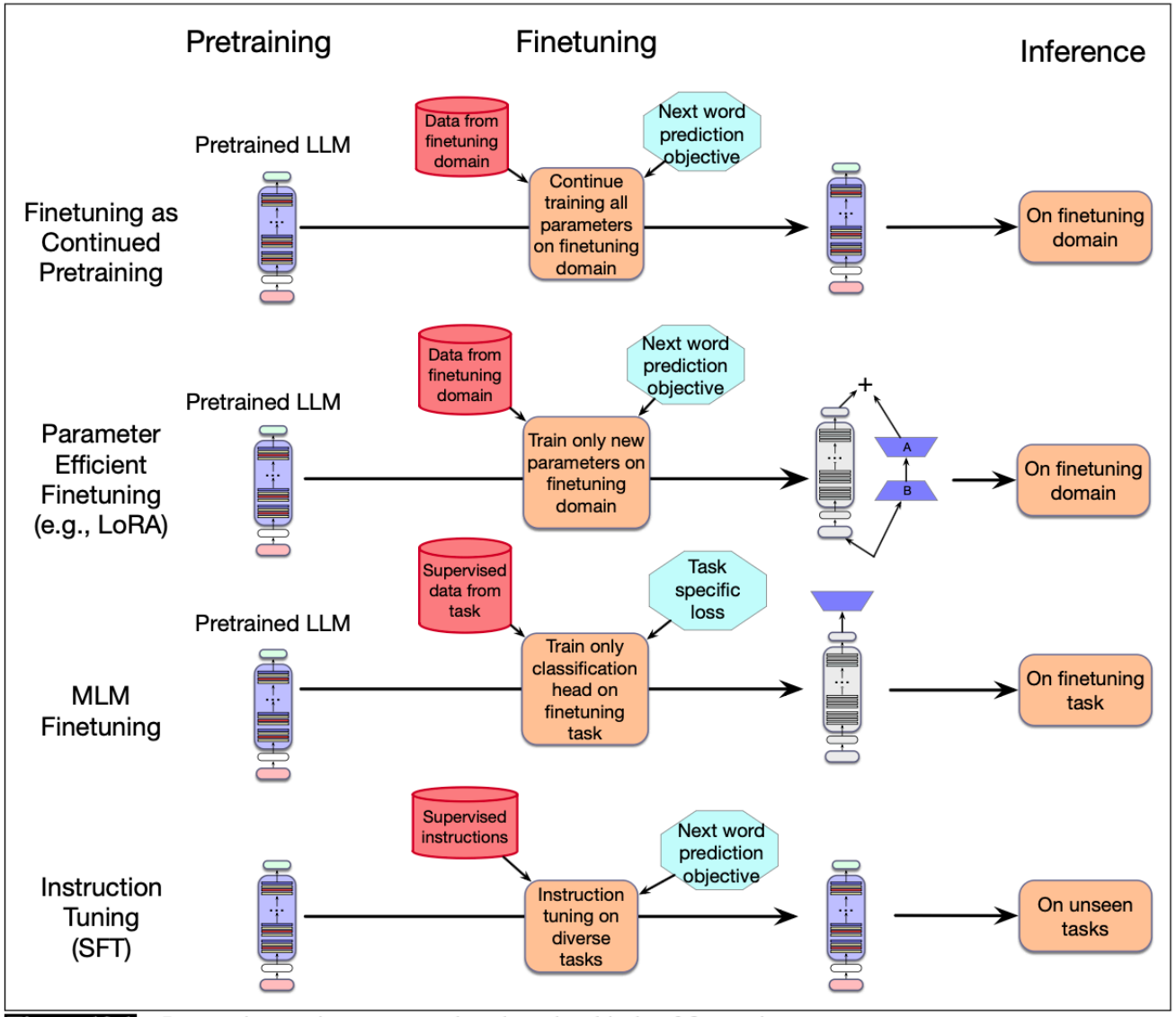

Figure 10.1 Instruction tuning compared to the other kinds of finetuning.

Instruction Tuning (SFT)

Finetune a pretrained model on instruction-response pairs
Still uses the next-token prediction objective
The model learns to follow instructions across many tasks

Instruction Examples from the Aya Dataset

por	Qual é a capital do estado da Bahia?	A capital do estado da Bahia, no Brasil, é Salvador. Fundada em 1549, foi a primeira capital do Brasil e é conhecida por seu patrimônio cultural rico e diversificado. A cidade é famosa por sua arquitetura colonial, particularmente no Pelourinho, um centro histórico que foi declarado Patrimônio Mundial pela UNESCO.
pes	در برج کبوترخانه میبد چند لانه کبوتر وجود دارد؟	کبوترخانه میبد برجی استوانه‌ای است که داخل آن چهارهزار لانه کبوتر وجود دارد و نوار سفید رنگی برای جذب کبوتران دورتادور برج کشیده شده است. این برج در برابر خطر حمله دیگر حیوانات به کبوترها بسیار ایمن بوده است.
msa	Apakah nasi lemak?	Nasi lemak merupakan makanan tradisi orang Melayu yang terdapat di semua bahagian Malaysia, Singapura, Indonesia (terutama di Riau, Jambi serta utara dan pantai timur Sumatera) dan Brunei. Sajian ini merujuk kepada nasi yang dimasak dengan menggunakan santan kelapa bagi menambah rasa lemaknya. Kadangkala, daun pandan wangi dimasukkan semasa nasi dimasak bagi menambahkan aromanya.
tam	செயற்கை நுண்ணறிவு என்றால் என்ன?	பொதுவாக மனிதர்களால் செய்யப்படும் பணிகளைச் செய்ய ஒரு கணினி அல்லது ஒரு கணினியால் கட்டுப்படுத்தப்படும் ஒரு ரோபோவின் திறன் செயற்கை நுண்ணறிவு எனப்படும்.

Figure 10.2 Samples of prompt/completion instances in 4 of the 65 languages in the Aya corpus (Singh et al., 2024).

Where Instruction Data Comes From

- Human-written instructions and demonstrations
- Existing NLP datasets converted into prompts
- Crowdsourcing guidelines reused as prompts
- Synthetic instructions generated by language models

Evaluating Instruction-Tuned Models

- Goal: generalize to unseen tasks
- Leave-one-task-cluster-out evaluation
- Held-out task families test true instruction following

Learning from Preferences

- Instruction tuning is not enough for alignment
- Preference learning teaches models what humans prefer
- Targets helpfulness, harmlessness, honesty, and correctness

Preference Data

- Humans compare alternative model outputs
- Data may come from annotators, social media votes, or other LLMs
- Chosen response vs rejected response creates supervision

Bradley-Terry Preference Model

- Preference probability is modeled using a logistic sigmoid
- The preferred output should receive a higher reward score
- Differences in reward scores determine preference probabilities

Reward Models

- A reward model predicts how good a response is Initialize from a pretrained LLM
- Replace the language-model head with a scalar reward head

Reward Model Loss

- Train with cross-entropy over chosen vs rejected responses
- The loss encourages preferred outputs to get higher scores
- This reward signal is later used for alignment

RLHF Pipeline

Step 1: Start with a pretrained base model
Step 2: Apply supervised finetuning (SFT)
Step 3: Train a reward model from preferences
Step 4: Optimize the aligned model with reinforcement learning

Direct Preference Optimization (DPO)

- Alternative to RLHF
- Optimizes preferences directly without explicit RL
- Often simpler and more stable than PPO-based RLHF

Test-Time Compute

- Extra computation at inference can improve quality
- Examples: chain-of-thought, self-consistency, tree search
- No retraining required

Chain-of-Thought Reasoning

- Prompt the model to reason step by step
- Improves arithmetic, logic, and planning tasks
- Encourages explicit intermediate reasoning

Tradeoffs in Post-training

- Better alignment improves helpfulness and safety
- More compute increases latency and cost
- Practical systems balance quality, speed, and reliability

Key Takeaway

- Pretraining creates a language model
- Post-training creates an assistant
- Instruction tuning, alignment, and test-time compute are all essential